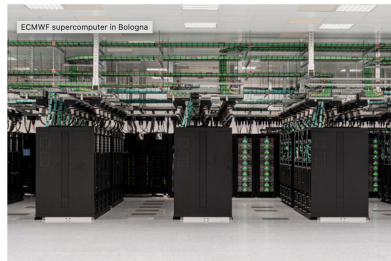


Tailoring Machine Learning Weather Predictions for Local Impacts

Ozma Houck, James Franke

What's Been Happening in Weather Forecasting?

- Traditional numerical weather prediction (NWP) models are expensive and slow to run.
 - Ex: Best global weather model created by European Centre for Medium-Range Weather Forecasts (ECMWF) requires a supercomputer.
- Beginning in 2022, several machine learning (ML) based weather models have been created that have rivaled or exceeded the accuracy of traditional models.
- Critically, these models can be run on a personal laptop.



The Centre's supercomputer facility is made up of four Atos BullSequana XH2000 clusters.

ECMWF Supercomputer

Increased Accessibility of Weather Predictions Opens New Opportunities

- Individual countries and smaller organizations can now run their own weather models instead of relying on large government agencies.
- Global models are more accurate in some areas than others.
- Different groups prioritize accuracy for different weather variables.

Increased Accessibility of Weather Predictions Opens New Opportunities

- Individual countries and smaller organizations can now run their own weather models instead of relying on large government agencies.
- Global models are more accurate in some areas than others.
- Different groups prioritize accuracy for different weather variables.
- **The Rub:** All global weather models (NWP or ML) are trained to optimize overall global accuracy.
 - A temperature error in Greenland is treated the same as a temperature error in Chicago!
 - The ML models might be cheap to run, but they are extremely expensive to train!
- **Potential Solution:** Post-process global model outputs with a small, localized model to improve accuracy for specific regions and weather variables of interest.

How to Post-Process Weather Model Outputs? What Outputs Matter Most?

- Just as ML tools can be used to post-process models just as they have been used to build them.
- **Contribution 1:** Demonstrate that a simple low-compute neural network model can meaningfully improve the accuracy of both traditional and ML weather models. Additionally, show where these simple models perform well, and where they struggle.
- **Contribution 2:** show that post-processing using alternative loss functions can lead to greater improvements for specific weather variables of interest.

Previous Post-Processing Papers Focused on Narrow Geographic Areas or Use Expensive Models

- Rasp 2018: Use neural network to post-process IFS forecasts in Germany. Use Neural network to train mean and standard deviation of temperature deviations.
- Han 2021: show how CNNs can be used to correct IFS on Gridded ERA5 data over China
- Trotta 2025: show that NWP and ML models can be post-processed in similar ways. Use process based model used by Australian Meterological Office

Fine-Tuning Approach

Data:

- Forecast predictions from IFS (NWP) and Pangu (ML) Global Models
- Ground truth data from ERA5 Reanalysis

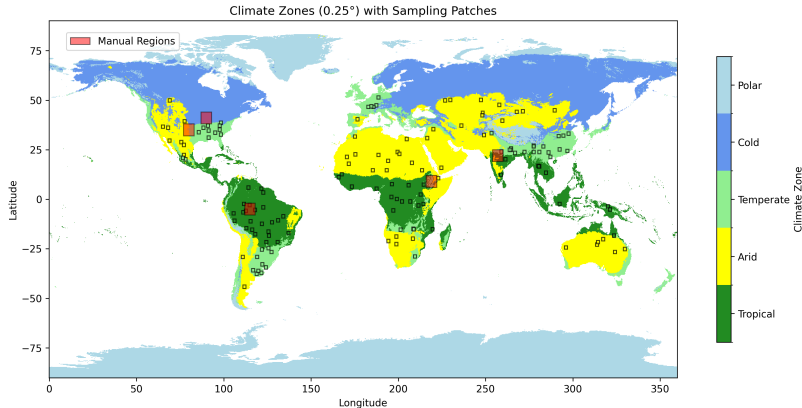
Model:

- Train a small neural network for a specific variable in a specific region.
- Loss Function: Mean Squared Error (MSE)
- Test simple multi layer perceptron (MLP) and convolution based UNET architectures.

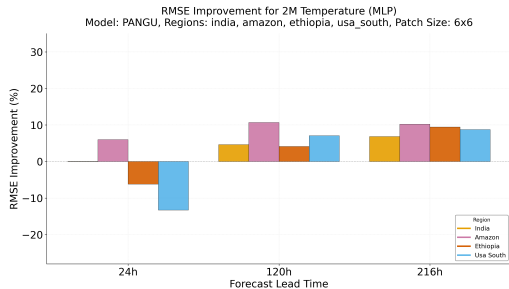
Training Details

Start by Manually Looking at Five Regions of Interest

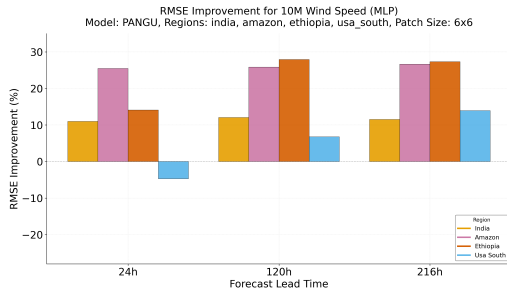
- Manually choose a few regions with a variety of climates and land terrains.
- Sample from different climate zones (arid, temperate, tropical).
- Sample from different topographic zones (coastal, mountainous, flat).



See Greater Improvements for Longer Lead Times and for Wind Speed

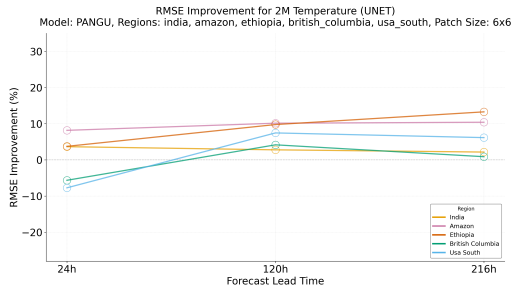


(a) 2m Temperature: Pangu (ML Model)

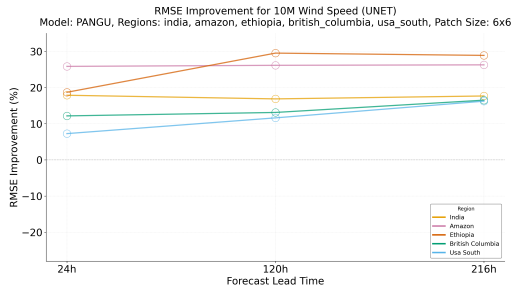


(b) Wind Speed: Pangu (ML Model)

Slightly Better Performance for UNet vs MLP

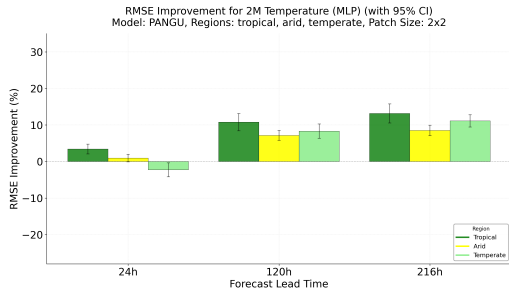


(a) 2m Temperature: Pangu (ML Model)

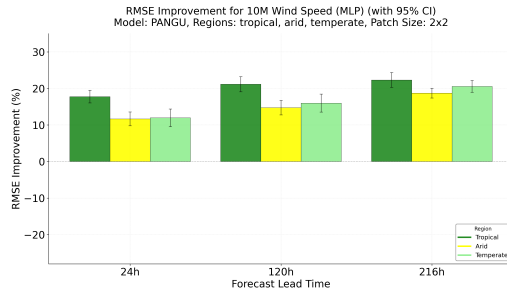


(b) Wind Speed: Pangu (ML Model) using UNet

Climate Zones

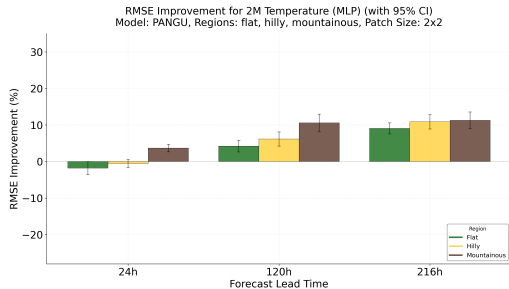


(a) 2m Temperature: Pangu

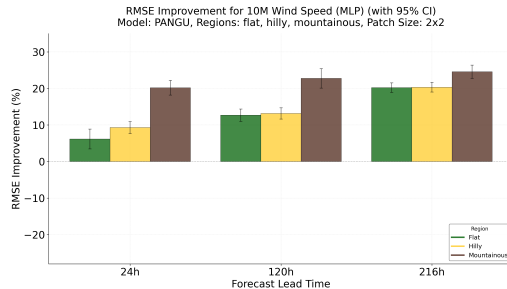


(b) Wind Speed: Pangu

Topographic Zones



(a) 2m Temperature: Pangu



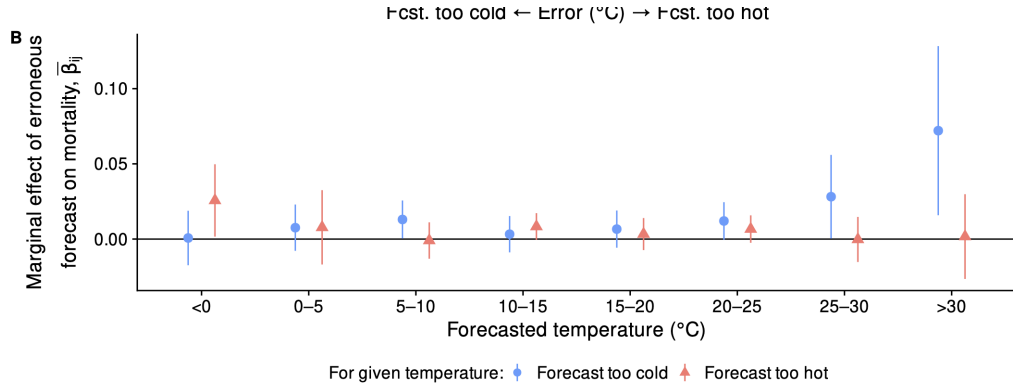
(b) Wind Speed: Pangu

All Deterministic Forecasts Trained with MSE, All Ensemble Models Trained with CRPS

- Know that under-prediction errors of extreme temperatures have a disproportionate impact on mortality rates.
- Temperature errors around freezing point have outsized impacts on agriculture
- Wind speed errors during high wind events have larger impacts on wind power generation than errors during calm periods.

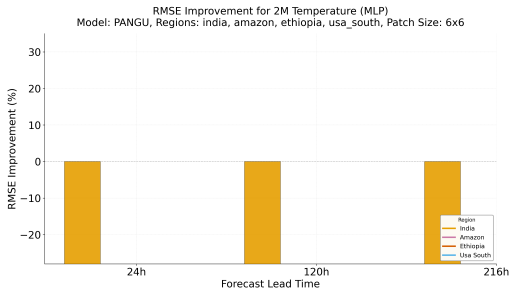
Question: If we know the damage function of different types of errors, can we use custom loss functions to meaningfully improve accuracy for these impactful errors while still maintaining overall accuracy?

Version 0: Damage Weighted MSE Loss Function



Mortality errors figure

Able to improve performance based on damage weighted MSE



Extreme heat trained model

pangu/lead_time/multi_region/6x6/lead

Training Details I

MLP Details:

- 2 hidden layers, 1024 nodes each
- ReLU activation
- Dropout rate ~ 0.1
- 16 dimensional lead time embeddings for 3 lead times.

UNet Details:

- Starting hidden dimension 128 (XX what does this mean)
- dropout_rate ~ 0.1
- 16 dimensional lead time embeddings for 3 lead times.

Training Details:

- Optimizer: Adam

Training Details II

- Learning Rate chosen using cosine annealing
- Weight decay $\sim 3e - 6$
- Hyperparameters chosen using bayesian hyperparameter optimization (hyperopt)

Data Details:

- Targets: ERA5 reanalysis (specify variables used, e.g., 2m temperature, 10m wind, total precipitation)
- Inputs: Global model outputs (Pangu / AIFS), regridded to region resolution
- Train / validation / test splits: e.g., train 2021-01-01–2023-12-31, test 2024-01-01–2024-12-31
- Sine and Cosine day of year

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